

Calculus 2

7 Lecture 7

Power Series

Definition 7.1. A functional series of the form

$$\sum_{k=0}^{\infty} a_k(x - x_0)^k = a_0 + a_1(x - x_0) + a_2(x - x_0)^2 + \cdots + a_k(x - x_0)^k + \cdots \quad (1)$$

is called a *power series centered at x_0* .

$x_0 \in \mathbb{R}$ is called the *center* of the series.

$a_k \in \mathbb{R}$, $k \in \mathbb{N} \cup \{0\}$ are the *coefficients* of the series.

Since by the linear substitution $t = x - x_0$, the power series (1) can be rewritten as a power series centered at 0:

$$\sum_{k=0}^{\infty} a_k t^k = a_0 + a_1 t + a_2 t^2 + \cdots + a_k t^k + \cdots, \quad (2)$$

without loss of generality, we can study only power series of the form (2).

Definition 7.2. Let

$$D = \left\{ x \in \mathbb{R} \mid \sum_{k=0}^{\infty} a_k x^k \text{ converges} \right\} \subseteq \mathbb{R}.$$

That is, D is the *set of all points of convergence* of $\sum_{k=0}^{\infty} a_k x^k$.

Then,

$$R = \sup\{|x| \mid x \in D\}$$

is called the *radius of convergence* of the power series $\sum_{k=0}^{\infty} a_k x^k$. The set D is called the *interval of convergence* of the power series $\sum_{k=0}^{\infty} a_k x^k$.

Note that, since $0 \in D$, we have $D \neq \emptyset$, which implies that $R = \sup\{|x| \mid x \in D\}$ exists.

Claim 7.1. Let $\sum_{k=0}^{\infty} a_k x^k$ be a power series and let R be its radius of convergence. Then:

1. $\forall x \in \mathbb{R} : |x| < R$, the series $\sum_{k=0}^{\infty} a_k x^k$ converges *absolutely*;
2. $\forall x \in \mathbb{R} : |x| > R$, the series $\sum_{k=0}^{\infty} a_k x^k$ *diverges*.

Proof. 1) Due to [Def 3.1](#), it suffices to show that the series

$$\sum_{k=0}^{\infty} |a_k x^k|$$

converges.

Since

$$|x| < R = \sup\{|y| \mid y \in D\}, \quad D = \left\{ y \in \mathbb{R} \mid \sum_{k=0}^{\infty} a_k y^k \text{ converges} \right\},$$

by the definition of the supremum, there exists $y \in D$ such that $|x| < |y|$. Since $y \in D$, the series $\sum_{k=0}^{\infty} a_k y^k$ converges, therefore, due to [Claim 1.5](#), we have

$$\lim_{k \rightarrow \infty} a_k y^k = 0,$$

which implies that

$$\exists M > 0 \forall k \in \mathbb{N} \cup \{0\} \rightarrow |a_k y^k| < M.$$

Therefore, for every $k \in \mathbb{N} \cup \{0\}$ we have

$$|a_k x^k| = |a_k y^k| \left| \frac{x}{y} \right|^k < M \left| \frac{x}{y} \right|^k = M q^k,$$

where

$$0 \leq |x| < |y| \implies q = \left| \frac{x}{y} \right| \in [0, 1).$$

The latter implies that the series

$$\sum_{k=0}^{\infty} M q^k$$

converges. Therefore, due to [Claim 5.7](#) [The Weierstrass M-test], the series

$$\sum_{k=0}^{\infty} |a_k x^k|$$

converges.

2) Assume that $|x| > R$ and $\sum_{k=0}^{\infty} a_k x^k$ converges. The latter implies (see [Def 7.2](#)) that $x \in D$. Therefore, since

$$R = \sup\{|y| \mid y \in D\},$$

we have $|x| \leq R$, which contradicts $|x| > R$. Thus, our assumption is false and the series $\sum_{k=0}^{\infty} a_k x^k$ diverges. $\square$

Claim 7.2 (The Cauchy–Hadamard Formula). For any power series $\sum_{k=0}^{\infty} a_k x^k$, its radius of convergence R is given by

$$\frac{1}{R} = \overline{\lim}_{k \rightarrow \infty} |a_k|^{1/k} = \lim_{k \rightarrow \infty} \sup_{n \geq k} |a_n|^{1/n}. \quad (*)$$

Proof. Recall that, by [Def 7.2](#), we have

$$D = \left\{ x \in \mathbb{R} \mid \sum_{k=0}^{\infty} a_k x^k \text{ converges} \right\} \quad \text{and} \quad R = \sup\{|x| \mid x \in D\}.$$

It is clear that $0 \in D$. Let $x \neq 0$, then we have

$$L = \overline{\lim}_{k \rightarrow \infty} |a_k x^k|^{1/k} = \overline{\lim}_{k \rightarrow \infty} |a_k|^{1/k} |x| = |x| \tau, \quad (1)$$

where

$$\tau = \overline{\lim}_{k \rightarrow \infty} |a_k|^{1/k}.$$

[Equality \(1\)](#) holds true since x does not depend on k and $|x| > 0$ (see [Claim 7.4](#)).

Then, due to [Claim 2.9](#) [The Root Test], the series $\sum_{k=0}^{\infty} |a_k x^k|$ converges if $L = |x| \tau < 1$ and diverges if $L = |x| \tau > 1$.

- If $\tau = 0$, then $|x| \tau = 0 < 1$, $\forall x \in \mathbb{R}$, thus $D = \mathbb{R}$ and $R = +\infty$; that is, [Formula \(*\)](#) holds true if $\tau = 0$.
- If $\tau = +\infty$, then $|x| \tau = +\infty > 1$, $\forall x \in \mathbb{R} \setminus \{0\}$, thus $D = \{0\}$ and $R = 0$; that is, [Formula \(*\)](#) holds true if $\tau = +\infty$.
- If $\tau \in (0, +\infty)$, then

$$|x| \tau < 1 \iff |x| < \frac{1}{\tau},$$

hence

$$\left\{ x \in \mathbb{R} \mid |x| < \frac{1}{\tau} \right\} \subseteq D,$$

which implies that $R \geq \frac{1}{\tau}$.

Let us assume that $R > \frac{1}{\tau}$. Then, due to [Def 7.2](#), there is $x \in D$ such that

$$R > |x| > \frac{1}{\tau}.$$

Inequality $|x| > \frac{1}{\tau}$ implies that the series $\sum_{k=0}^{\infty} |a_k x^k|$ diverges (since $L = |x|\tau > 1$), and inequality $R > |x|$ implies that the series $\sum_{k=0}^{\infty} a_k x^k$ converges (due to [Claim 7.1](#)). Thus, the assumption is false.

Therefore, $R = \frac{1}{\tau}$ and [Formula \(*\)](#) holds true if $\tau \in (0, +\infty)$. $\square$

Note that [Claim 7.1](#) tells nothing about the behavior of the series $\sum_{k=0}^{\infty} a_k x^k$ at the boundary points $x = R$ and $x = -R$, if $R \in (0, +\infty)$.

In fact, all logically admissible possibilities can occur:

- the series can be absolutely convergent at $x = R$ and $x = -R$;

for example:

$$\sum_{k=1}^{\infty} \frac{1}{k^2} x^k, \quad R = 1, \quad D = [-1, 1];$$

- the series can be divergent at $x = R$ and $x = -R$;

for example:

$$\sum_{k=1}^{\infty} \frac{1}{k} x^{2k}, \quad R = 1, \quad D = (-1, 1);$$

- the series can be divergent at $x = R$ and conditionally convergent at $x = -R$;

for example:

$$\sum_{k=1}^{\infty} \frac{1}{k} x^k, \quad R = 1, \quad D = [-1, 1);$$

- the series can be conditionally convergent at $x = R$ and divergent at $x = -R$;

for example:

$$\sum_{k=1}^{\infty} \frac{(-1)^k}{k} x^k, \quad R = 1, \quad D = (-1, 1].$$

Claim 7.3 (On Uniform Convergence of Power Series). Let $\sum_{k=0}^{\infty} a_k x^k$ be a power series with the radius of convergence $R > 0$. Then:

1. the series converges *uniformly* on every interval

$$I = [a, b] \subset (-R, R);$$

2. if the series *converges* at $x = R$, then, it converges *uniformly* on every interval

$$I = [a, R] \subset (-R, R];$$

3. if the series *diverges* at $x = R$, then, it converges *non-uniformly* on every interval

$$I = [a, R) \subset (-R, R).$$

Proof. 1) Let

$$m = \max\{|a|, |b|\}.$$

It is clear that $0 \leq m < R$. Note that:

- $\forall k \in \mathbb{N} \cup \{0\}, \forall x \in I = [a, b],$

$$|a_k x^k| \leq |a_k| m^k =: M_k;$$

- since $|m| < R$, due to [Item 1\)](#) of [Claim 7.1](#), the series

$$\sum_{k=0}^{\infty} M_k = \sum_{k=0}^{\infty} |a_k| m^k$$

converges.

Thus, due to [Claim 5.7](#) [The Weierstrass M-test], the series $\sum_{k=0}^{\infty} a_k x^k$ converges uniformly on $[a, b] \subset (-R, R)$.

2) Due to [Item 1\)](#) and [Claim 6.6](#), it suffices to prove the claim for $I = [0, R]$.

Let

$$f_k(x) = a_k R^k \quad \text{and} \quad g_k(x) = \left(\frac{x}{R}\right)^k, \quad x \in [0, R], \quad \forall k \in \mathbb{N} \cup \{0\}.$$

Note that:

- the series

$$\sum_{k=0}^{\infty} f_k(x) = \sum_{k=0}^{\infty} a_k R^k$$

converges uniformly on $[0, R]$.

Indeed, the series converges since this is one of the conditions of the claim; since the series $\sum_{k=0}^{\infty} a_k R^k$ does not effectively depend on x , we can say, due to [Def 5.3](#), that it converges uniformly on $[0, R]$.

- $\forall k \in \mathbb{N} \cup \{0\}, \forall x \in [0, R],$

$$\left(\frac{x}{R}\right)^k \geq \left(\frac{x}{R}\right)^{k+1},$$

i.e. $g_k(x) \geq g_{k+1}(x);$

- $\forall k \in \mathbb{N} \cup \{0\}, \forall x \in [0, R],$

$$\left|\frac{x}{R}\right| < 1,$$

i.e. $g_k(x) \downarrow 0.$

Thus, due to [Claim 5.9](#) [The Abel Test for the UC], the series

$$\sum_{k=0}^{\infty} f_k(x) g_k(x) = \sum_{k=0}^{\infty} a_k R^k \left(\frac{x}{R}\right)^k = \sum_{k=0}^{\infty} a_k x^k$$

converges uniformly on $[0, R]$.

3) The pointwise convergence of $\sum_{k=0}^{\infty} a_k x^k$ on $[a, R)$ follows from [Item 1\)](#) of [Claim 7.1](#). Let us assume that the series

$$\sum_{k=0}^{\infty} a_k x^k = \lim_{n \rightarrow \infty} f_n(x), \quad f_n(x) = \sum_{k=0}^n a_k x^k,$$

converges uniformly on $[a, R)$. Note that:

- $\lambda = R$ is a finite limit point of the set $[a, R)$;

- $\forall n \in \mathbb{N}$, we have

$$\lim_{x \rightarrow R} f_n(x) = \sum_{k=0}^n a_k R^k \in \mathbb{R}$$

(since n is finite).

Thus, due to [Item 1](#)) of [Claim 6.1](#) [On Changing the Order of Two Limits], the limit

$$\lim_{n \rightarrow \infty} \sum_{k=0}^n a_k R^k = \sum_{k=0}^{\infty} a_k R^k$$

exists. Since this contradicts the conditions of the claim, our assumption is false and the series $\sum_{k=0}^{\infty} a_k x^k$ converges non-uniformly on $[a, R) \subset (-R, R)$. $\square$

Claim 7.4. Let $\{a_k\}_{k=1}^{\infty}$ and $\{b_k\}_{k=1}^{\infty}$ be two numerical sequences. If

$$\exists \lim_{k \rightarrow \infty} a_k = \alpha > 0,$$

then

$$\overline{\lim}_{k \rightarrow \infty} (a_k b_k) = \alpha \cdot \overline{\lim}_{k \rightarrow \infty} b_k.$$

Proof. See Calculus-1. $\square$

Claim 7.5 (On Derivative of Power Series). Let

$$S(x) = \sum_{k=0}^{\infty} a_k x^k$$

on $(-R, R)$, where $R > 0$ is the radius of convergence of the power series. Then:

1. the radius of convergence of the series

$$\sum_{k=1}^{\infty} k a_k x^{k-1}$$

is equal to R ;

2. for every $x \in (-R, R)$, we have

$$S'(x) = \left(\sum_{k=0}^{\infty} a_k x^k \right)' = \sum_{k=1}^{\infty} k a_k x^{k-1}. \quad (3)$$

Proof. 1) Note that:

- the series

$$\sum_{k=1}^{\infty} k a_k x^{k-1} \quad \text{and} \quad \sum_{k=1}^{\infty} k a_k x^k = x \sum_{k=1}^{\infty} k a_k x^{k-1}$$

have the same interval of convergence, therefore, due to [Def 7.2](#), they have the same radius of convergence;

- since

$$\lim_{k \rightarrow \infty} k^{1/k} = 1 > 0,$$

due to [Claim 7.4](#), we have the equality

$$\begin{aligned} \overline{\lim}_{k \rightarrow \infty} |k a_k|^{1/k} &= \overline{\lim}_{k \rightarrow \infty} \left(k^{1/k} |a_k|^{1/k} \right) \\ &= 1 \cdot \overline{\lim}_{k \rightarrow \infty} |a_k|^{1/k} = \overline{\lim}_{k \rightarrow \infty} |a_k|^{1/k}. \end{aligned}$$

which, due to [Claim 7.2](#) [The Cauchy–Hadamard Formula], implies that the series $\sum_{k=1}^{\infty} k a_k x^{k-1}$ and $\sum_{k=0}^{\infty} a_k x^k$ have the same radius of convergence.

Therefore, the series $\sum_{k=1}^{\infty} k a_k x^{k-1}$ and $\sum_{k=0}^{\infty} a_k x^k$ have the same radius of convergence and [Item 1](#)) is proved.

2) Let $x_0 \in (-R, R)$ be arbitrary. Then it ought to be clear that there is an open interval $J = (-r, r)$ such that $x_0 \in J$ and $0 < r < R$.

Let

$$u_k(x) = a_k x^k, \quad \forall k \in \mathbb{N} \cup \{0\},$$

and

$$f_n(x) = \sum_{k=0}^n u_k(x), \quad \forall n \in \mathbb{N} \cup \{0\}.$$

Then note that:

- a) $\forall k \in \mathbb{N} \cup \{0\}$, $u_k(x) = a_k x^k$ is differentiable on $J = (-r, r)$;
- b) the series $\sum_{k=0}^{\infty} u_k(x)$ converges at $x = 0 \in J$;
- c) the series

$$\sum_{k=0}^{\infty} u'_k(x) = \sum_{k=1}^{\infty} k a_k x^{k-1}$$

converges uniformly on J .

Indeed, due to [Item 1](#)), the radius of convergence of the series $\sum_{k=1}^{\infty} k a_k x^{k-1}$ is equal to $R > 0$. Therefore, due to [Item 1](#)) of [Claim 7.3](#), it converges uniformly on the interval $[-r, r] \subset (-R, R)$, which implies the uniform convergence of the series on $(-r, r)$.

That is, all conditions of [Claim 6.13](#) [Term-by-term Diff. of Series] are satisfied and [Equality \(3\)](#) holds true $\forall x \in (-r, r)$ due to [Item 2](#)) of [Claim 6.13](#). Then, since $x_0 \in (-R, R)$ is arbitrary, [Equality \(3\)](#) holds true $\forall x \in (-R, R)$ and [Item 2](#)) is proved. $\square$

Note that the series $\sum_{k=0}^{\infty} a_k x^k$ and $\sum_{k=1}^{\infty} k a_k x^{k-1}$ have the same radius of convergence, but the series may have different intervals of convergence (see [Def 7.2](#)).

For example, if $a_0 = 0$, $a_k = \frac{1}{k^2}$, $\forall k \in \mathbb{N}$, we have

$$\sum_{k=0}^{\infty} a_k x^k = \sum_{k=1}^{\infty} \frac{1}{k^2} x^k \quad \text{with } R = 1 \text{ and } D = [-1, 1],$$

and

$$\sum_{k=1}^{\infty} k a_k x^{k-1} = \sum_{k=1}^{\infty} \frac{1}{k} x^{k-1} \quad \text{with } R = 1 \text{ and } D = [-1, 1).$$

Claim 7.6 (Analyticity of Power Series). Let

$$S(x) = \sum_{k=0}^{\infty} a_k x^k$$

on $(-R, R)$, where $R > 0$ is the radius of convergence of the series. Then $S \in C^\infty(-R, R)$ and, for $\forall n \in \mathbb{N}$, $\forall x \in (-R, R)$, we have

$$S^{(n)}(x) = \sum_{k=n}^{\infty} k(k-1) \cdots (k-n+1) a_k x^{k-n},$$

where $C^\infty(-R, R)$ is the set of all infinitely differentiable functions on $(-R, R)$ and $S^{(n)}(x)$ is the n -th derivative of $S(x)$.

Proof. Follows directly from [Claim 7.5](#) via mathematical induction on $n \in \mathbb{N}$. $\square$

Claim 7.7 (On Antiderivative of Power Series). Let

$$S(x) = \sum_{k=0}^{\infty} a_k x^k$$

on $(-R, R)$, where $R > 0$ is the radius of convergence of the series. Then:

1. the radius of convergence of the series

$$\sum_{k=0}^{\infty} \frac{a_k}{k+1} x^{k+1}$$

is equal to R ;

2. for every $x \in (-R, R)$, we have

$$\int_0^x S(t) dt = \int_0^x \left(\sum_{k=0}^{\infty} a_k t^k \right) dt = \sum_{k=0}^{\infty} \frac{a_k}{k+1} x^{k+1} = \sum_{k=0}^{\infty} \left(\int_0^x a_k t^k dt \right). \quad (4)$$

Proof. 1) Note that:

- the series

$$\sum_{k=0}^{\infty} \frac{a_k}{k+1} x^{k+1} = x \sum_{k=0}^{\infty} \frac{a_k}{k+1} x^k$$

and the series

$$\sum_{k=0}^{\infty} \frac{a_k}{k+1} x^k$$

have the same interval of convergence, therefore, due to [Def 7.2](#), they have the same radius of convergence;

- since

$$\lim_{k \rightarrow \infty} \left(\frac{1}{k+1} \right)^{1/k} = 1 > 0,$$

due to [Claim 7.4](#), we have the equality

$$\begin{aligned} \overline{\lim}_{k \rightarrow \infty} \left| \frac{1}{k+1} a_k \right|^{1/k} &= \overline{\lim}_{k \rightarrow \infty} \left(\left(\frac{1}{k+1} \right)^{1/k} |a_k|^{1/k} \right) \\ &= 1 \cdot \overline{\lim}_{k \rightarrow \infty} |a_k|^{1/k} = \overline{\lim}_{k \rightarrow \infty} |a_k|^{1/k}. \end{aligned}$$

which, due to [Claim 7.2](#) [The Cauchy–Hadamard Formula], implies that the series $\sum_{k=0}^{\infty} \frac{a_k}{k+1} x^k$ and $\sum_{k=0}^{\infty} a_k x^k$ have the same radius of convergence.

Therefore, the series $\sum_{k=0}^{\infty} \frac{a_k}{k+1} x^{k+1}$ and $\sum_{k=0}^{\infty} a_k x^k$ have the same radius of convergence and [Item 1\)](#) is proved.

- 2) Let $x \in (-R, R)$, $I = [0, x] \subset (-R, R)$,

$$u_k(t) = a_k t^k, \quad \forall k \in \mathbb{N} \cup \{0\},$$

and

$$f_n(t) = \sum_{k=0}^n u_k(t).$$

Then note that:

- a) $\forall k \in \mathbb{N} \cup \{0\}$, $u_k(t) = a_k t^k$ is Riemann integrable on $I = [0, x]$;

b) the series $\sum_{k=0}^{\infty} a_k t^k$ converges uniformly on $I = [0, x]$.

The latter holds since $I = [0, x] \subset (-R, R)$ and due to [Item 1](#)) of [Claim 7.3](#). Thus, all conditions of [Claim 6.12](#) [Term-by-term Int. of Series] are satisfied and, for every $x \in (-R, R)$, we have

$$\int_0^x S(t) dt = \int_0^x \left(\sum_{k=0}^{\infty} a_k t^k \right) dt \stackrel{1}{=} \sum_{k=0}^{\infty} \left(\int_0^x a_k t^k dt \right) \stackrel{2}{=} \sum_{k=0}^{\infty} \frac{a_k}{k+1} x^{k+1}.$$

Here, 1) is due to [Claim 6.12](#) and 2) follows from

$$\int_0^x a_k t^k dt = \frac{a_k}{k+1} t^{k+1} \Big|_0^x = \frac{a_k}{k+1} x^{k+1}.$$

Thus, [Equality \(4\)](#) is proved. $\square$

Note that [Item 1](#)) of [Claim 7.7](#) tells us that the series $\sum_{k=0}^{\infty} a_k x^k$ and $\sum_{k=0}^{\infty} \frac{a_k}{k+1} x^{k+1}$ have the same radius of convergence, but the series may have different intervals of convergence (see [Def 7.2](#)).

For example, if $a_0 = 0$, $a_k = \frac{1}{k}$, $\forall k \in \mathbb{N}$, we have

$$\sum_{k=0}^{\infty} a_k x^k = \sum_{k=1}^{\infty} \frac{1}{k} x^k \quad \text{with } R = 1 \text{ and } D = [-1, 1),$$

and

$$\sum_{k=0}^{\infty} \frac{a_k}{k+1} x^{k+1} = \sum_{k=1}^{\infty} \frac{1}{k(k+1)} x^{k+1} \quad \text{with } R = 1 \text{ and } D = [-1, 1].$$

Claim 7.8 (On Equality of Power Series). Let

$$f(x) = \sum_{k=0}^{\infty} a_k x^k \quad \text{on } (-R_1, R_1)$$

and

$$g(x) = \sum_{k=0}^{\infty} b_k x^k \quad \text{on } (-R_2, R_2),$$

where $R_1 > 0$ and $R_2 > 0$ are radii of convergence of the power series $\sum_{k=0}^{\infty} a_k x^k$ and $\sum_{k=0}^{\infty} b_k x^k$ respectively. If there is a numerical sequence $\{x_n\}_{n=1}^{\infty}$ such that:

a) $x_n \in (-r, r) \setminus \{0\}$, $\forall n \in \mathbb{N}$, where

$$r = \min\{R_1, R_2\} > 0;$$

b) $\lim_{n \rightarrow \infty} x_n = 0$;

c) $f(x_n) = g(x_n)$ for every $n \in \mathbb{N}$,

then $a_k = b_k$, $\forall k \in \mathbb{N} \cup \{0\}$.

Proof. Let

$$c_k = a_k - b_k, \quad \forall k \in \mathbb{N} \cup \{0\}.$$

Then, for every $x \in (-r, r)$, we have

$$h(x) := \sum_{k=0}^{\infty} c_k x^k = \sum_{k=0}^{\infty} (a_k - b_k) x^k \stackrel{1}{=} \sum_{k=0}^{\infty} a_k x^k - \sum_{k=0}^{\infty} b_k x^k = f(x) - g(x), \quad (**)$$

where [Equality 1](#)) holds since both series converge due to [Item 1](#)) of [Claim 7.1](#).

From [Formula \(**\)](#) it follows that:

- the series

$$h(x) = \sum_{k=0}^{\infty} c_k x^k$$

has the radius of convergence $R \geq r > 0$;

•

$$h(x_n) = 0, \quad \forall n \in \mathbb{N}; \tag{5}$$

- it suffices to prove that

$$c_k := a_k - b_k = 0, \quad \forall k \in \mathbb{N} \cup \{0\}.$$

Let us assume that there is $k \in \mathbb{N} \cup \{0\}$ such that $c_k \neq 0$. Then

$$m = \min\{k \in \mathbb{N} \cup \{0\} \mid c_k \neq 0\}$$

is well-defined and, for every $x \in (-r, r)$, we have

$$\begin{aligned} h(x) &= \sum_{k=0}^{\infty} c_k x^k = \sum_{k=0}^{m-1} c_k x^k + \sum_{k=m}^{\infty} c_k x^k \\ &= 0 + \sum_{k=m}^{\infty} c_k x^k = x^m \sum_{k=0}^{\infty} c_{k+m} x^k. \end{aligned}$$

Here, the first finite sum is 0 since $c_k = 0$ for every $k \in \{0, 1, \dots, m-1\}$.

It ought to be clear that the series

$$\widehat{h}(x) := \sum_{k=0}^{\infty} c_{k+m} x^k$$

and

$$h(x) = x^m \sum_{k=0}^{\infty} c_{m+k} x^k = x^m \widehat{h}(x)$$

have the same interval of convergence, therefore, due to [Def 7.2](#), the series have the same radius of convergence $R \geq r > 0$.

Since, due to [Item a\)](#), $x_n \neq 0$ for every $n \in \mathbb{N}$, the equalities

$$h(x) = x^m \widehat{h}(x), \quad \forall x \in (-r, r),$$

and [Formula \(5\)](#) imply that

$$\widehat{h}(x_n) = 0, \quad \forall n \in \mathbb{N}.$$

Now note that

$$c_m \stackrel{1}{=} \sum_{k=0}^{\infty} c_{k+m} 0^k \stackrel{2}{=} \widehat{h}(0) \stackrel{3}{=} \widehat{h}\left(\lim_{n \rightarrow \infty} x_n\right) \stackrel{4}{=} \lim_{n \rightarrow \infty} \widehat{h}(x_n) \stackrel{5}{=} \lim_{n \rightarrow \infty} 0 = 0.$$

Indeed:

1. since $0^0 := 1$ and $0^k = 0$, $\forall k \in \mathbb{N}$;
2. by definition, $\widehat{h}(x) = \sum_{k=0}^{\infty} c_{m+k} x^k$, $\forall x \in (-r, r)$, $r > 0$;
3. $\lim_{n \rightarrow \infty} x_n = 0$ due to [Item b\)](#);

4. due to [Item 2](#) of [Claim 7.5](#), $\hat{h}$ is continuous on $(-r, r)$; since $\lim_{n \rightarrow \infty} x_n = 0 \in (-r, r)$, this implies that

$$\lim_{n \rightarrow \infty} \hat{h}(x_n) = \hat{h}\left(\lim_{n \rightarrow \infty} x_n\right);$$

5. due to [Equality \(5\)](#), we have $\hat{h}(x_n) = 0, \forall n \in \mathbb{N}$.

That is, we arrived at a contradiction (with $c_m \neq 0$). Therefore, the assumption is false and $c_k = 0, \forall k \in \mathbb{N} \cup \{0\}$. $\square$

Now we can easily prove [Claim 4.2](#) [The Abel Theorem on the Cauchy product].

Claim 7.9 (The Abel Theorem on the Cauchy product). Let $\sum_{k=0}^{\infty} a_k$ and $\sum_{k=0}^{\infty} b_k$ be two convergent series with sums a and b respectively. If their Cauchy product $\sum_{k=0}^{\infty} c_k$, where

$$c_k = a_0 b_k + a_1 b_{k-1} + \cdots + a_k b_0, \quad \forall k \in \mathbb{N} \cup \{0\},$$

is a convergent series with the sum c , then $c = a \cdot b$.

Proof. Let us consider the series:

$$f(x) := \sum_{k=0}^{\infty} a_k x^k, \quad g(x) := \sum_{k=0}^{\infty} b_k x^k, \quad h(x) := \sum_{k=0}^{\infty} c_k x^k.$$

Let R_1, R_2 , and R_3 be the radii of convergence of $f(x), g(x)$, and $h(x)$ respectively. Since

$$f(1) = \sum_{k=0}^{\infty} a_k 1^k = a, \quad g(1) = \sum_{k=0}^{\infty} b_k 1^k = b, \quad h(1) = \sum_{k=0}^{\infty} c_k 1^k = c,$$

due to [Def 7.2](#), we have $R_1 \geq 1, R_2 \geq 1$, and $R_3 \geq 1$. Therefore, due to [Item 1](#) (if $R_i > 1$) or [Item 2](#) (if $R_i = 1$) of [Claim 7.3](#), the series $f(x), g(x)$, and $h(x)$ converge uniformly on the interval $[0, 1]$.

Therefore, due to [Claim 6.11](#) [Continuity of a Series], the functions $f(x), g(x)$, and $h(x)$ are continuous on $[0, 1]$ and we have

$$\lim_{x \rightarrow 1-0} f(x) \stackrel{1}{=} f(1) = a, \quad \lim_{x \rightarrow 1-0} g(x) \stackrel{1}{=} g(1) = b, \quad \lim_{x \rightarrow 1-0} h(x) \stackrel{1}{=} h(1) = c. \quad (6)$$

[Equalities 1](#)) hold since the functions are continuous at $x = 1$.

Since $R_1 \geq 1$ and $R_2 \geq 1$, due to [Item 1](#) of [Claim 7.1](#), the series $f(x) = \sum_{k=0}^{\infty} a_k x^k$ and $g(x) = \sum_{k=0}^{\infty} b_k x^k$ converge absolutely for every $x \in [0, 1)$.

Note that, for every $k \in \mathbb{N} \cup \{0\}, \forall x \in \mathbb{R}$, we have

$$c_k x^k = (a_0 x^0)(b_k x^k) + (a_1 x^1)(b_{k-1} x^{k-1}) + \cdots + (a_k x^k)(b_0 x^0).$$

That is, due to [Def 4.1](#),

$$h(x) = \sum_{k=0}^{\infty} c_k x^k$$

is the Cauchy product of the series

$$\sum_{k=0}^{\infty} a_k x^k = f(x) \quad \text{and} \quad \sum_{k=0}^{\infty} b_k x^k = g(x).$$

Therefore, due to [Claim 4.1](#) [The Mertens Theorem], we have

$$h(x) = f(x) \cdot g(x), \quad \text{for every } x \in [0, 1). \quad (7)$$

To finish the proof, note that:

$$c \stackrel{1}{=} \lim_{x \rightarrow 1-0} h(x) \stackrel{2}{=} \lim_{x \rightarrow 1-0} (f(x)g(x)) \stackrel{3}{=} \left(\lim_{x \rightarrow 1-0} f(x) \right) \left(\lim_{x \rightarrow 1-0} g(x) \right) \stackrel{4}{=} ab.$$

Here, 1) and 4) are due to [Equalities \(6\)](#), 2) is due to [Equality \(7\)](#), and 3) is due to the [Equality \(7\)](#) for every $x < 1$ and we can assume that $x > 0$. $\square$

Claim 7.10. Let $\sum_{k=0}^{\infty} a_k x^k$ be a power series with the radius of convergence $R > 0$. Then

$$\forall p \in (0, R) \quad \exists C > 0 \quad \forall k \in \mathbb{N} \cup \{0\} \quad \rightarrow \quad |a_k| \leq \frac{C}{p^k}.$$

Proof. Since $p \in (0, R)$, due to [Item 1\)](#) of [Claim 7.1](#), the series

$$\sum_{k=0}^{\infty} |a_k| p^k$$

converges. Therefore, due to [Claim 1.5](#),

$$\lim_{k \rightarrow \infty} |a_k| p^k = 0.$$

It is well-known (see Calculus-1) that every convergent sequence is bounded. Thus we have

$$\exists C > 0 \quad \forall k \in \mathbb{N} \cup \{0\} \quad \rightarrow \quad |a_k| p^k < C.$$

Since $p > 0$, the claim is proved. $\square$